

MyMavenGuavaApp

Step1:

To create a MyMavenGuavaApp project, run the following command:

```
mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
```

```
vaishnavi@vaishnavi-virtualbox:~/devops21$ mvn archetype:generate -DgroupId=com.example -DartifactId=MyMavenGuavaApp1 -DarchetypeArtifactId=maven-archetype-quickstart -DinteractiveMode=false
[INFO] Scanning for projects...
[INFO] ----- org.apache.maven:standalone-pom -----
[INFO] Building Maven Stub Project (No POM) 1
[INFO] -----[ pom ]-----
[INFO] >>> maven-archetype-plugin:3.4.1:generate (default-cli) > generate-sources @ standalone-pom >>>
[INFO] <<< maven-archetype-plugin:3.4.1:generate (default-cli) < generate-sources @ standalone-pom <<<
[INFO] --- maven-archetype-plugin:3.4.1:generate (default-cli) @ standalone-pom ---
[INFO] Generating project in Batch mode
[INFO] No archetype defined. Using maven-archetype-quickstart (org.apache.maven.archetypes:maven-archetype-quickstart:1.0)
[INFO] Using following parameters for creating project from Old (1.x) Archetype: maven-archetype-quickstart:1.0
[INFO] -----
[INFO] Parameter: basedir, Value: /home/vaishnavi/devops21
[INFO] Parameter: package, Value: com.example
[INFO] Parameter: groupId, Value: com.example
[INFO] Parameter: artifactId, Value: MyMavenGuavaApp1
[INFO] Parameter: packageName, Value: com.example
[INFO] Parameter: version, Value: 1.0-SNAPSHOT
[INFO] Project created from Old (1.x) Archetype in dir: /home/vaishnavi/devops21/MyMavenGuavaApp1
[INFO] -----
[INFO] BUILD SUCCESS
[INFO] -----
[INFO] Total time: 0.742 s
[INFO] Finished at: 2026-03-24T22:00:09+05:30
[INFO] -----
[1] Killed gedit pom.xml (wd: ~/devops21/MyMavenApp)
(wd now: ~/devops21)
vaishnavi@vaishnavi-virtualbox:~/devops21$
```

The command can be broken down as:

`DgroupId=com.example` – project's base package (reverse-domain style).

`-DartifactId=mymavenguavaapp` – defines the name of the project (will also be the folder name).

`-DarchetypeArtifactId=maven-archetype-quickstart` – sets the Maven template to be the basic Java application.

`-DinteractiveMode=false` – deactivates interactive mode and uses the given values automatically.

Step 2:

Use `$cd` command to go into the directory.

`$gedit pom.xml`

`pom.xml` file is created to configure the Maven project. It defines the project information, adds required dependencies like Guava and Commons IO libraries, and includes build plugins to compile the Java code, run unit tests, and package the application into an executable JAR file.

Add the dependencies:

```
<!-- Source: https://mvnrepository.com/artifact/com.google.guava/guava -->
```

```
<dependency>
```

```

<groupId>com.google.guava</groupId>

<artifactId>guava</artifactId>

<version>33.5.0-jre</version>

<scope>compile</scope>
</dependency>

<!-- Source: https://mvnrepository.com/artifact/commons-io/commons-io -->
<dependency>

  <groupId>commons-io</groupId>

  <artifactId>commons-io</artifactId>

  <version>2.21.0</version>

  <scope>compile</scope>
</dependency>

```

Execute \$tree command to view the structure of the project

```

vaishnavi@vaishnavi-virtualbox:~/devops21$ cd myMavenGuavaApp1
vaishnavi@vaishnavi-virtualbox:~/devops21/PyHavenGuavaApp1$ tree
.
├── pom.xml
├── src
│   ├── main
│   │   ├── java
│   │   │   └── com
│   │   │       └── example
│   │   │           └── App.java
│   └── test
│       ├── java
│       │   └── com
│       │       └── example
│       │           └── AppTest.java
└── 9 directories, 3 files

```

Step 3:

Edit the App.java file

This step opens the App.java file using the gedit editor. The code is pasted to create a Java program that uses the Google Guava library to create an immutable list and Apache Commons IO library to copy a file from source to destination

```
package com.example;
```

```

/**
 * Hello world!
 *

```

```
*/  
// Import FileUtils class from Apache Commons IO library  
// This is used to perform file operations like copying files  
import org.apache.commons.io.FileUtils;  
  
// Import File class to represent file and directory pathnames  
import java.io.File;  
  
// Import IOException to handle input-output errors  
import java.io.IOException;  
  
// Import ImmutableList from Google Guava library  
// ImmutableList creates a list that cannot be modified after creation  
import com.google.common.collect.ImmutableList;  
  
// Define the main class  
public class App  
{  
    // Main method - program execution starts from here  
    public static void main( String[] args )  
    {  
        // Create an immutable list of strings (fruits)  
        // This list cannot be changed (no add/remove allowed)  
        ImmutableList<String> fruits = ImmutableList.of("Apple", "Banana", "Cherry");  
  
        // Print the list of fruits to the console  
        System.out.println(fruits);  
  
        // Create a File object representing the source file  
        File sourceFile = new File("source.txt");
```

```
// Create a File object representing the destination file
File destFile = new File("destination.txt");

try {
    // Copy the contents of source.txt to destination.txt
    // Using Apache Commons IO FileUtils class
    FileUtils.copyFile(sourceFile, destFile);

    // Print success message if file is copied
    System.out.println("File copied successfully!");
}
catch (IOException e) {
    // If any error occurs during file copying,
    // this block will handle the exception

    // Print error message
    System.err.println("Error occurred while copying file: " + e.getMessage());
}
}
```

Step 4:

Execute the command `$mvn compile,test,package`

If the command shoes output as “build success”, it means that the project has compiled successfully

```

vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$ mvn compile
[INFO] Scanning for projects...
[INFO]
[INFO] -----[ com.example.MyMavenGuavaApp ]-----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.426 s
[INFO] Finished at: 2026-03-24T22:57:20+05:30
[INFO]
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$

vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$ src$ cd ..
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$ mvn test
[INFO] Scanning for projects...
[INFO]
[INFO] -----[ com.example.MyMavenGuavaApp ]-----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/MyMavenGuavaApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenGuavaApp ---
[INFO]
[INFO] -----[ T E S T S ]-----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.029 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 1.525 s
[INFO] Finished at: 2026-03-24T22:57:41+05:30
[INFO]
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$

[INFO] -----[ com.example.MyMavenGuavaApp ]-----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO]
[INFO] --- maven-resources-plugin:2.6:resources (default-resources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/MyMavenGuavaApp/src/main/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:compile (default-compile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-resources-plugin:2.6:testResources (default-testResources) @ MyMavenGuavaApp ---
[WARNING] Using platform encoding (UTF-8 actually) to copy filtered resources, i.e. build is platform dependent!
[INFO] skip non existing resourceDirectory /home/vaishnavi/devops21/MyMavenGuavaApp/src/test/resources
[INFO]
[INFO] --- maven-compiler-plugin:3.1:testCompile (default-testCompile) @ MyMavenGuavaApp ---
[INFO] Nothing to compile - all classes are up to date
[INFO]
[INFO] --- maven-surefire-plugin:2.22.2:test (default-test) @ MyMavenGuavaApp ---
[INFO]
[INFO] -----[ T E S T S ]-----
[INFO] Running com.example.AppTest
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0, Time elapsed: 0.003 s - in com.example.AppTest
[INFO]
[INFO] Results:
[INFO]
[INFO] Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
[INFO]
[INFO] --- maven-jar-plugin:3.1.2:jar (default-jar) @ MyMavenGuavaApp ---
[INFO]
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 3.422 s
[INFO] Finished at: 2026-03-24T22:57:59+05:30
[INFO]
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$

```

Step 5:

Under the MyMavenGuavaApp project directory, create a file named source.txt and add some sample text such as “Hello Class”. This file will act as the source file that the Java program copies to destination.txt during execution.

Step 6:

Execute `$ mvn exec:java -Dexec.mainClass="com.example.App"`

```
[INFO] -----
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$ mvn exec:java -Dexec.mainClass="com.example.App"
[INFO] Scanning for projects...
[INFO]
[INFO] ----- com.example:MyMavenGuavaApp -----
[INFO] Building MyMavenGuavaApp 1.0-SNAPSHOT
[INFO] -----[ jar ]-----
[INFO]
[INFO] --- exec-maven-plugin:3.0.3:java (default-cli) @ MyMavenGuavaApp ---
[INFO] Apple, Banana, Cherry
[INFO] File copied successfully!
[INFO] -----
[INFO] BUILD SUCCESS
[INFO]
[INFO] Total time: 0.456 s
[INFO] Finished at: 2026-03-24T23:00:43+05:30
[INFO] -----
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$
```

Step 7:

After executing the program, the `destination.txt` file will be automatically created inside the `MyMavenGuavaApp` directory. The command `cat destination.txt` is used to display the contents of the file and verify the output.

```
[INFO] -----
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$ ls
destination.txt pom.xml source.txt src target
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$ cat destination.txt
hiiiiiiiiiiiiiiii
vaishnavi@vaishnavi-virtualbox:~/devops21/MyMavenGuavaApp$
```

Pushing The Project into Git:

Execute the commands:

`$ git config --global user.name "vaishnavim1121"`

`$ git config --global user.email "vaishnavim1121@gmail.com"`

`$ git init`

`$ git add .`

`$ git commit -m "initial commit"`

`$ git remote add origin https://github.com/vaishnavim1121/MyMavenGuavaApp.git`

`$ git remote set-url origin https://*PAT*@github.com/vaishnavim1121/MyMavenGuavaApp`

`$ git push -u origin master`

```

Activities Terminal Mar 17 10:27
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03

Note: If your new repositories, which will suppress this warning, call:
Note:
Note: git config --global init.defaultBranch <name>
Note:
Note: Names commonly chosen instead of 'master' are 'main', 'trunk' and
Note: 'development'. The just-created branch can be renamed via this command:
Note:
Note: git branch -m <name>
Initialized empty Git repository in /home/vaishnavi/devops21/MyMavenSeleniumApp03/.git/
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $ git add -
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $ git commit -m "first commit"
[master (root-commit) 1af4a8b] first commit
11 files changed, 285 insertions(+)
create mode 100644 pom.xml
create mode 100644 src/main/java/com/example/App.java
create mode 100644 src/test/java/com/example/AppTest.java
create mode 100644 target/MyMavenSeleniumApp03-1.0-SNAPSHOT.jar
create mode 100644 target/classes/com/example/App.class
create mode 100644 target/maven-archiver/pom.properties
create mode 100644 target/maven-status/maven-compiler-plugin/compile/default-compile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/createdFiles.lst
create mode 100644 target/maven-status/maven-compiler-plugin/testCompile/default-testCompile/inputFiles.lst
create mode 100644 target/surefire-reports/TEST-com.example.AppTest.txt
create mode 100644 target/surefire-reports/com.example.AppTest.txt
create mode 100644 target/test-classes/com/example/AppTest.class
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $ git config --global user.name "vaishnavin121"
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $ git config --global user.email "vaishnavin121@gmail.com"
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $ git remote add origin https://github.com/vaishnavin121/MyMavenSeleniumApp03.git
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $ set-url origin https://ghp_hmYmPp55xgt2QxFOV7noGjtJal6ZV3wVf7I@github.com/vaishnavin121/MySeleniumApp03
set-url: command not found
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $ git remote set-url origin https://ghp_hmYmPp55xgt2QxFOV7noGjtJal6ZV3wVf7I@github.com/vaishnavin121/MySeleniumApp03
remote: Repository not found.
fatal: repository 'https://github.com/vaishnavin121/MySeleniumApp03/' not found
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $ git remote set-url origin https://ghp_hmYmPp55xgt2QxFOV7noGjtJal6ZV3wVf7I@github.com/vaishnavin121/MyMavenSeleniumApp03
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $ git push -u origin master
Enumerating objects: 39, done.
Counting objects: 100% (39/39), done.
Delta compression using up to 16 threads
Compressing objects: 100% (19/19), done.
Writing objects: 100% (39/39), 8.46 KiB | 721.00 KiB/s, done.
Total 39 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/vaishnavin121/MyMavenSeleniumApp03
 * [new branch] master -> master
Branch 'master' set up to track remote branch 'master' from 'origin'.
vaishnavi@vaishnavi-VirtualBox: ~/devops21/MyMavenSeleniumApp03 $

```